Continue developing GhostRoute. First, read the existing project and preserve all working features. Do not rewrite working code unnecessarily.

1. PERMANENT REQUIREMENTS

Create or update "PROJECT_REQUIREMENTS.md" in the repository root. Record all requirements, features, links, architecture decisions, and completed tasks there. Update it whenever requirements change or features are implemented.

Also update "AGENTS.md" with instructions to read "PROJECT_REQUIREMENTS.md" at the beginning of every session and before starting new tasks.

2. ABOUT / CREDITS PAGE — REQUIRED

The app keeps losing its About links. Fix this permanently.

Create a proper About/Credits screen accessible from Settings, with functional, clickable external links:

- Developer: Amirreza
- GitHub: https://github.com/amrsh88
- Telegram Channel: https://t.me/net_republic
- Telegram Community: https://t.me/netrepublic

Identify Net Republic as a technology and free/open-internet community.

Requirements:

- Every link must open the correct destination.
- Use Android-compatible external link intents.
- Handle missing browser or Telegram apps gracefully.
- Keep the links visible and accessible.
- Use the existing design system.
- Add an app version and license information only when accurate.
- No fake links or placeholder buttons.

Add automated tests where practical. Verify the screen is reachable from Settings and all links are correct.

3. FEATURES TO IMPLEMENT AND TRACK

Location simulation

- Route Studio with route drawing, editing, and waypoints.

- Route playback with pause, resume, seeking, speed adjustment, looping, and ping-pong.
- Floating joystick for manual movement.
- Location profiles with saved configurations.
- Altitude, bearing, and accuracy simulation.
- GPX/KML/CSV import and export.
- Location recording and replay.

Maps and navigation

- Map styles and search improvements.
- Search using Photon and maps using the existing MapLibre/OpenFreeMap architecture.
- Saved locations and coordinates.
- Offline-map support where technically feasible.
- Optional road-following and routing.

Automation

- Tasker integration and Android intents.

- Quick Settings tile and useful shortcuts.
- Scheduled simulation where supported.

Diagnostics and reliability

- Session dashboard and useful simulation statistics.
- Clear error messages and recovery controls.
- Diagnostics for location-provider failures and lost mock-location permission.
- Backup and restore of user configurations.
- Test important functionality and handle lifecycle interruptions correctly.

UI and experience

The current UI is already good. Improve it rather than replacing it:

- Add only 1–2 complementary accent colors.

- Smooth animations and transitions.
- Better loading, success, error, and empty states.
- Consistent spacing, typography, and visual hierarchy.
- Keep everything responsive and avoid unnecessary animations that hurt performance.

Privacy

GhostRoute should remain privacy-first and offline-first wherever possible. No advertisements, analytics, or account requirements.

4. NEW FEATURE: PUBLIC IP / VPN DETECTION

The existing IP button currently shows my real nearby location instead of reliably identifying the VPN's public IP. Fix the actual underlying issue.

- Detect the externally observed public IPv4 and IPv6 addresses.
- Display the detected IP, geolocation, provider/ISP when available, and last-updated timestamp.
- Determine whether requests are actually routed through the VPN, rather than assuming a VPN is active because Android reports a connection.
- Handle split tunneling, unavailable services, IPv6 leaks, timeouts, and stale results honestly.
- Allow manual refresh and configurable periodic refresh, with a minimum interval of 60 seconds.
- Show clear loading, error, and stale-data states.
- Fix the existing IP history and synchronization functionality.
- Keep IP history persistent across app restarts.
- Sync simulated GPS with the detected IP location only when explicitly enabled.
- Never silently overwrite a manually selected location.

Do not fabricate IP addresses, locations, VPN status, or provider information.

5. IMPLEMENTATION RULES

- Inspect the existing architecture before modifying anything.
- Maintain compatibility with the existing Android implementation.
- Write tests for new logic and regression cases.
- Run the relevant Gradle tests and build after changes.
- Do not break existing working functionality.
- Update PROJECT_REQUIREMENTS.md with completed tasks, remaining work, test results, and architectural decisions.
- Keep commits small and meaningful.
- If a feature cannot be implemented reliably, document the blocker instead of pretending it

works.

6. EXECUTION ORDER

1. Inspect the repository and current project requirements.
2. Fix and verify the About/Credits page and all four identity/link entries.
3. Fix public IP detection, VPN routing verification, and persistent IP history.
4. Improve UI animations and polish.
5. Implement the remaining features incrementally, prioritizing reliability and usefulness.
6. Run tests, verify the build, and update the project requirements after each completed task.

Start by inspecting the current state. Report what already exists, what is broken, and your first implementation step. Do not blindly

existing features.GHOSTROUTE: ADDITIONAL FEATURE ROADMAP

Update PROJECT_REQUIREMENTS.md with every feature below. Mark each as PLANNED, IN PROGRESS, COMPLETE, or BLOCKED. Inspect the existing implementation first and avoid duplicating features.

1. ADVANCED ROUTE STUDIO

- Multi-stop routes with draggable waypoints.
- Route reversal and automatic loop closure.
- Route templates: walking, cycling, driving, train, and custom.
- Configurable acceleration and deceleration.
- Smooth transitions between stationary and moving states.
- Per-segment speed and movement profiles.
- Randomized but bounded GPS accuracy and

bearing.
- Realistic stoplights, traffic stops, and pauses.
- Elevation profiles and altitude interpolation.
- Route preview with total distance, estimated duration, and waypoint count.
- Undo/redo, duplicate route, and route history.
- Route bookmarks and favorites.
- Export and import routes as GPX and KML.

2. REALISTIC LOCATION ENGINE

- Configurable GPS update intervals.
- Smooth interpolation between coordinates.
- Configurable horizontal accuracy.
- Realistic bearing changes during turns.
- Optional speed and altitude simulation.
- Configurable stationary-location drift.
- Seeded randomness for reproducible tests.
- Boundary-aware movement and route constraints.

- Robust handling of zero-length segments and duplicate coordinates.
- Correct behavior at route endpoints and during rapid seeking.
- A simulation clock that remains consistent when playback is paused or resumed.
- A diagnostics screen showing generated location values and update timing.

Do not claim to simulate actual satellite signals. This is software-level location simulation.

3. LIVE MAP EXPERIENCE

- Search places and coordinates directly from the map.
- Long-press to select a destination.
- Display current simulated coordinates.
- Display distance and bearing to the selected destination.

- Coordinate-format conversion: decimal degrees, DMS, and UTM where practical.
- Copy and share coordinates.
- Save named locations.
- Display route distance and estimated travel time.
- Map layer controls and customizable map styles.
- Optional compass and scale indicator.
- A recenter button and follow-location mode.
- Cache map data where the map provider and licensing permit.
- Gracefully handle unavailable tiles and network failures.

4. LOCATION PROFILES

- Save multiple simulation configurations.
- Assign custom names, icons, and descriptions.
- Store routes, speeds, accuracy settings, and

- Import and export profile backups.
- Restore the last active profile after restarting the app.
- Detect invalid or incompatible imported configurations.
- Include example profiles for testing and demonstration.

5. IP INTELLIGENCE DASHBOARD

Improve the existing IP button into a useful network diagnostics dashboard.

Display:

- Public IPv4 and IPv6.
- Network type and available connectivity information.
- VPN connection status as reported by Android.
- Independently verified externally observed IP

addresses.
- Approximate public-IP geolocation.
- ISP or network organization when available.
- DNS resolver information where reliably obtainable.
- Last successful lookup and data freshness.
- IPv4/IPv6 connectivity status.
- Errors, timeouts, and unavailable services.
- IP history with timestamps and change detection.
- Comparison between consecutive network states.

Important:

- Do not confuse the device's GPS location with public-IP geolocation.
- Do not assume a connected VPN routes every request through its tunnel.
- Clearly distinguish verified observations from

- Keep diagnostic requests minimal and disclose external services used.

6. IP HISTORY AND NETWORK MONITORING

- Persistent history using Room or the existing persistence architecture.
- Detect public-IP changes.
- Show a timeline of network transitions.
- Record whether IPv4 or IPv6 was available.
- Allow manual snapshots.
- Filter and search history.
- Delete individual records or clear history.
- Export selected records to CSV or JSON.
- Configurable monitoring with a minimum interval of 60 seconds.
- Pause monitoring when disabled by the user.
- Avoid duplicate records and unnecessary background network requests.

7. AUTOMATION AND QUICK ACCESS

- Quick Settings tile to start or stop an existing simulation.
- Home-screen shortcuts for saved profiles.
- Android App Shortcuts where appropriate.
- Intent-based integration with Tasker and automation apps.
- Documented intent actions and parameter validation.
- Optional scheduled start/stop functionality, respecting Android background restrictions.
- Notifications showing simulation status and providing pause/stop controls.
- Persistent notifications only while required by the active foreground service.

8. SIMULATION SESSION DASHBOARD

Create a session screen containing:

- Elapsed simulation time.
- Current coordinates.
- Current speed and bearing.
- Route progress.
- Distance traveled.
- Remaining distance and estimated time.
- Current profile and playback mode.
- Last location update.
- Active simulation status.
- Start, pause, resume, stop, and restart controls.

All displayed values must come from actual engine state, not hardcoded demo values.

9. DATA MANAGEMENT

- Automatic persistence of user preferences.
- Save and restore the last active route and profile.

- Versioned backup format.
- Import validation and useful error messages.
- Export all user-created data.
- Optional encrypted backup files using Android-supported cryptography.
- Schema migrations that preserve existing user data.
- Clear cache separately from user data.
- A reset-settings action with explicit confirmation.

10. UI AND INTERACTION

The existing UI is already good. Improve it without rebuilding everything.

Add:

- Consistent accent colors and semantic status colors.

- Smooth screen transitions.
- Animated route progress.
- Subtle map-selection animations.
- Responsive layouts for different screen sizes.
- Better empty states and loading indicators.
- Haptic feedback where appropriate.
- Accessible touch targets.
- Dark and light themes.
- Reduced-motion support.
- Search history and recent destinations.
- Contextual tooltips for complicated controls.
- Clear error messages with recovery actions.

Avoid excessive gradients, meaningless animations, fake statistics, and decorative controls that do nothing.

11. SECURITY AND PRIVACY

- No accounts or mandatory cloud services.

- No advertisements or unnecessary analytics.
- No location transmission without explicit user action.
- Explain when external map, geolocation, or IP services receive requests.
- Use Android's supported permission and foreground-service mechanisms.
- Never collect credentials or unrelated personal information.
- Keep logs free of unnecessary sensitive data.
- Allow users to clear locally stored history.
- Do not promise complete anonymity or guaranteed VPN leak prevention.

12. TESTING AND RELIABILITY

- Unit tests for route interpolation and movement.
- Tests for wrap, reflection, endpoint clamping, and seeking.
- Tests for invalid coordinates and malformed

imports.
- Persistence and migration tests.
- Service lifecycle and permission-loss tests.
- Tests for IP lookup failures, stale data, and address-family differences.
- UI tests for About links and navigation.
- Regression tests for every bug fixed.
- Build and test on the project's existing Android configuration.
- Test real-device behavior where emulator tests are insufficient.

13. DEVELOPER EXPERIENCE

- Add a developer diagnostics page.
- Show app version, build variant, and relevant system information.
- Provide an exportable diagnostic report with sensitive information excluded by default.
- Document architecture and important modules.

- Document external dependencies and licenses.
- Maintain a changelog.
- Keep a persistent list of known issues and unfinished features.
- Add a regression checklist for releases.

IMPLEMENTATION PRIORITIES

P0: Preserve existing functionality, About links, reliable location simulation, service lifecycle, and persistent configuration.

P1: Fix public-IP detection, IP history, network diagnostics, and session dashboard.

P2: Route Studio improvements, location profiles, GPX/KML support, and map interaction.

P3: Quick Settings tile, automation integration, backup/export, and UI polish.

P4: Offline maps, advanced routing, optional encrypted backups, and other technically demanding features.

Before implementing each feature:

1. Inspect the existing code.
2. Identify dependencies and Android compatibility constraints.
3. Write a short implementation plan.
4. Implement the smallest complete increment.
5. Run relevant tests and build checks.
6. Update PROJECT_REQUIREMENTS.md and the changelog.
7. Commit verified work.

Do not dispatch unnecessary parallel agents. Do not claim a feature is complete until its implementation has been inspected and verified.

Start with P0 and P1. Report the existing state, the first task, and its acceptance criteria.